

Tom Stesco

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PROGRAMMING

EXPERT

Python

PROFICIENT

C++ • SQL • Bash

PRIOR EXPERIENCE

Java • Javascript • Go

TOOLS

PyTorch • vLLM

Airflow • Apache Spark

MySQL • BigQuery • Datastore

Kafka • PubSub • Protobuf

Docker • Kubernetes • Terraform

GCP • AWS • Azure

GNU/Linux • Git

EDUCATION

ETH ZURICH

MSc INTEGRATED BUILDING

SYSTEMS

Excellence Scholar

Oct 2018 | Zurich, Switzerland

UNIVERSITY OF WATERLOO

BASc HONOURS CIVIL ENGINEERING

Merit Scholarship

Jun 2016 | Waterloo, Canada

COURSEWORK

GRADUATE

Model Predictive Control

Mathematical Optimization

Computational Physics

Building Control and Automation

Computational Fluid Dynamics

Renewable Energy Technologies

Technology and Innovation Management

Innovation Leadership

Social Networks Research

UNDERGRADUATE

Software Engineering Design Project

Building Science

Bioprocess Engineering

Advanced Calculus

Probability and Statistics

Linear Algebra

EXPERIENCE

TENSTORRENT

Apr 2023 – present (1 year, 6 months) | Toronto, Canada

Senior Staff Field Application Engineer - AI/ML Software

Sep 2024 – present

- Developed Docker containerization toolkit for AI/ML applications on AI accelerator hardware and reference LLM inference server backend.
- Integrated, ran, and reported LLM evaluations on AI accelerator hardware to ensure model accuracy.

Staff Field Application Engineer - AI/ML Software

Apr 2023 – Sep 2024

- Led development and deployment of AI application demos, proof-of-concepts, and sales tools in partnership with engineering team and contractors.
- Benchmarked transformer and CNN models on AI accelerator to track progress on performance and accuracy.
- Customer-facing engineering support to adapt transformer and CNN models for deployment on AI accelerator hardware.

BCG X

May 2022 – Apr 2023 (1 year) | Toronto, Canada

Senior AI/ML Engineer

May 2022 – Apr 2023

- Led MLOps for production pricing models in large (\$10+ billion/yr) eCommerce systems: code review (team of 5), test planning and validation, CI/CD, release management, monitoring, and on-call.
- Partnered with ML Scientists in developing ETL, ML model (XGBoost) training, and model deployment pipelines using Spark, Airflow, and Databricks Lakehouse.

ECOBEE

Oct 2018 – May 2022 (3 years, 8 months) | Toronto, Canada

Senior Data Scientist

Jun 2021 – May 2022

- Developed edge IoT device optimal control application (embedded C++) using ML models, resulting in proof of concept and patent application.

Data Scientist

Oct 2018 – Jun 2021

- Developed HVAC controls physics-based simulation platform, partially open sourced as `building-controls-simulator`, and was presented at the eSim2020 conference.
- Scaled internal IoT experimentation platform from 0 to 150k+ devices running parameterized experiments by automating build, deployment, monitoring, and metrics with C++, Python, PubSub, and Protobuf.

RESEARCH

AUTOMATIC CONTROL LAB

Sep 2017 – Oct 2018 (1 year, 1 month) | Zurich, Switzerland

Master's Student

Sep 2017 – Oct 2018

- Worked with Prof. Dr. John Lygeros and Dr. Annika Eichler on Human-in-the-Loop building controls project combining Model Predictive Control and online Bayesian ML.
- Implemented and tested proposed methods in literature for thermal comfort and occupancy prediction: Bayesian Linear Discriminant Analysis (LDA), Markov Chain Monte Carlo (MCMC), and change point detection algorithms.
- Developed Python library to train models and run controls simulation experiments on Euler HPC cluster.